Speaker Notes

Privacy-Preserving Human Activity Recognition from mmWave Radar Point Clouds using Lightweight Point Transformers
Team 10 · 23I006 Deep Learning · PSG College of Technology

Presenters: Abarajitha S K (23I301), G Aswath Rupan (23I321), Indu Vadana M D (23I323), Murugan S (23I338). Each slide lists its presenter, what to say, and likely examiner questions with prepared answers. Deck: Team10_mmWave_HAR.pptx, 27 slides; about 1.5 minutes per slide.

Slide 1: Privacy-Preserving Human Activity Recognition from mmWave Radar Point Clouds using Lightweight Point Transformers

Presenter: Abarajitha S K

Opening. We built a camera-free activity-recognition system that works on radar point clouds. Say the one-sentence result: our lightweight model (117.5 K parameters) reaches 96.83% on RadHAR and 49.56% on 49 MiliPoint actions performed by people it has never seen, and we show that how the data are split changes published accuracy dramatically. Order of presenters: Abarajitha (motivation, model), Aswath (data, preprocessing, noise), Indu (leakage, training, results), Murugan (robustness, ablation, demo, conclusion).

Possible examiner questions:

Q: *Why radar instead of a camera?*

> **A:** Radar gives only sparse 3-D points and velocities, never an image, so a person cannot be visually identified; it also works in the dark.

Q: *What is a point cloud here?*

> **A:** For each radar frame, a small set of detections (x, y, z, Doppler velocity, intensity), typically 6 to 21 per frame.

Slide 2: Radar senses activity without ever forming an image

Presenter: Abarajitha S K

Elderly monitoring and smart homes need continuous activity awareness. Cameras are accurate but intrusive and fail in the dark; wearables need compliance. Single-chip FMCW radars (TI IWR1443/1843, 60–81 GHz) output a sparse point cloud with a velocity per point. No image is ever formed, which is why the approach is privacy-preserving by construction. Note: we do not claim fall detection or clinical validity; neither dataset has fall labels.

Possible examiner questions:

Q: *Is radar data really private?*

> **A:** It carries body position and motion only; there is no face or texture. Gait could in principle identify people, which we mention as a limitation.

Q: *Which frequencies?*

A: IWR1443 and IWR1843 operate at 76–81 GHz.

Slide 3: Radar point clouds are sparse, noisy and temporal

Presenter: Abarajitha S K

Four difficulties define the problem. Sparsity: the median RadHAR frame has 21 returns and MiliPoint only 6. Noise: about a third of RadHAR returns are antenna echoes and half of the rest is static clutter. Time: the activity is in how the cloud moves, so a single frame is ambiguous. Evaluation: windows overlap heavily, so careless splits leak test data. The task: given a 2-second window, predict the activity.

Possible examiner questions:

Q: *Why 2-second windows?*

A: It is the RadHAR protocol (60 frames) and long enough to contain a repetition of each activity; we use the same 2 s on MiliPoint (48 frames).

Slide 4: Motivation: the same dataset, 34 % or 98 %?

Presenter: Abarajitha S K

On MiliPoint (49 actions) the dataset authors report 34.45% top-1 with PointNet++, while a 2025 paper reports 98.25% with a graph network under a random 80/10/10 split. Our leakage audit shows why: with random splitting of overlapping windows, 72.90% of the test frames already appear inside training windows. A model that is accurate only because of leakage is useless in a new home. This motivates two things: a model that is efficient and robust, and an evaluation that is honest.

Possible examiner questions:

Q: *Are you saying the 98% paper is wrong?*

A: We say its protocol measures something different: recognising near-duplicate windows of seen sequences, not generalising to new people. We reproduce the effect with our own models (protocol slide).

Slide 5: Published results use incompatible protocols

Presenter: Abarajitha S K

Representative work: voxel CNN-LSTM (RadHAR 2019), graph networks (MMPoint-GNN, Lee & Kim with joint pre-training, MAK-GCN), PointNet-LSTM hybrids (Gu et al. 2024), occupancy-gated convolution (OG-PCL 2025), FPGA-quantised pillars (Lim et al. 2024). Dense point-cloud backbones (PointNet++, DGCNN, Point Transformer, PTv3) were designed for LiDAR/CAD with thousands of points. Accuracies are as reported by each paper under its own protocol; they are not directly comparable, which is part of our point.

Possible examiner questions:

Q: *How many papers did you review?*

A: About 40 references, 19 from 2022–2026; the full list is in the paper and report.

Slide 6: Gap: radar structure and honest evaluation are under-used

Presenter: Abarajitha S K

Gap: radar-specific structure (sparsity, Doppler, static clutter) is rarely exploited explicitly; protocols are inconsistent and often leaky; robustness, transfer and edge cost are rarely reported together. Objectives map one-to-one to the contributions: a factorised point transformer with a motion descriptor; Doppler-based clutter removal; leakage-free protocols and an automated audit; full evaluation with seeds, significance, ablation, robustness, transfer and explainability; and a working demo.

Possible examiner questions:

Q: *What exactly is novel?*

> **A:** The factorised intra-/inter-segment attention for radar windows, the radar motion descriptor, Doppler-based person centring with gating, and the quantified leakage study. Attention, KD and quantisation themselves are established.

Slide 7: Two complementary radar datasets, used separately

Presenter: G Aswath Rupan

MiliPoint (primary): TI IWR1843 at 24 Hz, 11 subjects, 49 actions; 212,920 labelled frames, which matches the original paper exactly, so our loader is verified. Only xyz per point. RadHAR (secondary): TI IWR1443 at 30 Hz, 5 activities, 200 recordings (71.5 min train, 21.4 min test, matching the official partition), xyz plus Doppler and intensity. We do not merge the datasets: label sets, sensors and features differ, and MiliPoint action names are not published. We use the other dataset only for representation transfer, with harmonised axes and xyz input.

Possible examiner questions:

Q: *Why not merge the two datasets?*

> **A:** Different labels (5 vs 49, no mapping), different radars, and MiliPoint lacks Doppler. We run independent experiments and test transfer instead.

Q: *Licences?*

> **A:** RadHAR BSD-3-Clause, MiliPoint MIT.

Slide 8: Measured: very few points, clutter at fixed ranges

Presenter: G Aswath Rupan

Left: RadHAR. Panel (a) points per frame; (b) Doppler is quantised to a few values; (c) density in range–height: the person is at 1–2 m and there are static clutter bands at about 2.5 m and 4.5 m. MiliPoint is even sparser (median 6 points). Panel (c) of the MiliPoint figure shows the subject-disjoint split. Everything here is measured, not assumed.

Possible examiner questions:

Q: *How did you find the clutter?*

> **A:** Range histograms of all points versus moving (non-zero Doppler) points: the moving points sit only on the person; the far clusters never move.

Slide 9: Doppler-based gating removes static clutter

Presenter: G Aswath Rupan

Three steps. (1) ROI: drop antenna self-returns at x < 0.3 m (34.6% of returns). (2) Person centring: the median of the moving returns, because static objects have zero radial velocity. A plain centroid was pulled metres off the person. (3) Gating: keep returns within 1.2 m horizontally of the person; this removes 50.1% of the remaining returns, which are static clutter. Then 12 segments × 32 points per 2-s window. Intensity statistics come from training data only.

Possible examiner questions:

Q: *Could gating remove body parts?*

> **A:** Only returns more than 1.2 m from the body centre are removed; arm span and walking in place stay inside. The ablation slide shows the effect of removing gating.

Q: *Why not DBSCAN?*

> **A:** Clutter forms far, static clusters; a single interpretable radius around the moving centre removes it without per-window clustering cost or extra parameters.

Slide 10: Noise: handled in pre-processing vs. injected for testing

Presenter: G Aswath Rupan

We separate noise we measured in the data (handled by pre-processing) from synthetic perturbations we inject into the test set to probe robustness. The injected conditions are experiments, not claims about the datasets. Each condition has three severities and a fixed seed, so every model sees the identical perturbed test windows.

Possible examiner questions:

Q: *Why synthetic noise?*

> **A:** There is no ground truth of which returns are noise, so controlled perturbations are the only way to measure degradation systematically.

Slide 11: Our splits are clean; random splits leak most test frames

Presenter: Indu Vadana M D

Splits: RadHAR official test recordings unchanged; validation = 20% of training recordings (not windows), capped at 55 windows each. MiliPoint primary split is subject-disjoint (test subjects 8–10). The automated audit checks recording, subject and session overlap, identical frames, frames shared between overlapping windows, train-only statistics and checkpoint selection. Findings: our splits are clean. RadHAR's official test recordings were made minutes after training recordings on the same days and there are no subject IDs, so RadHAR cannot test new people; MiliPoint subject-disjoint is therefore our primary result. Random window splits leak 72.90% of test frames.

Possible examiner questions:

Q: *How do you know your test set was never used for tuning?*

> **A:** Checkpoints are chosen on validation macro-F1, the exported seed is chosen on validation, and the audit verifies checkpoint = argmax validation F1 for every run.

Q: *Why cap validation windows?*

A: One 186-s recording made up 20% of validation and was consistently misread; it dominated model selection. The fix was based on validation behaviour only.

Slide 12: STPT-Lite: factorised attention in 118 K parameters

Presenter: Abarajitha S K

Spatio-Temporal Point Transformer, Lite. Step 1: a shared MLP embeds every point. Step 2: self-attention only among the 32 points of each 5-frame segment, with a learned 3-D relative-position bias. Step 3: attentive plus max pooling gives one token per segment; we add an embedding of an 8-number motion descriptor (centroid, its displacement, mean Doppler, point count). Step 4: a [CLS] token and two Transformer blocks attend across the 12 segment tokens. Because attention is factorised, cost is 12×32^2 instead of 384^2 pairs, and there is no farthest-point sampling or k-NN. 117.5 K parameters, 36.60 MFLOPs. A 725 K-parameter teacher is distilled into it, and weights are stored in INT8.

Possible examiner questions:

Q: *Why factorise attention?*

A: Radar frames are too sparse for rich spatial structure but very regular in time; attending inside short segments and then across time keeps cost linear in the number of segments and avoids FPS/k-NN, which are slow on edge processors.

Q: *What does the motion descriptor add?*

A: Explicit kinematics that radar measures directly; its contribution is measured in the ablation.

Slide 13: Six baselines re-trained under one identical protocol

Presenter: Abarajitha S K

We re-implemented six baselines in pure PyTorch and trained all of them with identical data, augmentation, optimiser, epochs and early stopping: the RadHAR voxel CNN-LSTM, PointNet, PointNet+BiLSTM, PointNet++, DGCNN and Point Transformer. Our family: STPT-Large (teacher), STPT-Lite and STPT-Tiny, plus distillation and noise-robust training variants. Selection criteria from the brief: accuracy, robustness, generalisation, stability, size and latency.

Possible examiner questions:

Q: *Why not use PyTorch Geometric?*

A: It needs compiled extensions; pure PyTorch runs anywhere, including the CPU-only environment used here.

Q: *Is the comparison fair?*

A: Yes: same splits, pre-processing, optimiser, epoch budget, early stopping and seeds for every model.

Slide 14: Early stopping on held-out recordings and people

Presenter: Indu Vadana M D

AdamW, one-cycle learning rate (peak 2e-3), weight decay 0.05, label smoothing 0.1, gradient clipping 1.0, batch 64, at most 15 epochs on RadHAR and 30 on MiliPoint with early stopping (patience 5) on validation macro-F1; the best validation checkpoint is kept and the test set is evaluated once. Three seeds per configuration. The curves on the right are validation macro-F1 per epoch for seed 0 of each model; full histories (training/validation loss and accuracy, learning rate) are saved for every run.

Possible examiner questions:

Q: *How do you know you did not overfit?*

A: Early stopping on a recording- or subject-disjoint validation set, regularisation, small models, and agreement between validation and test accuracy.

Slide 15: RadHAR is saturated: all models reach 95–98 %

Presenter: Indu Vadana M D

Official RadHAR test recordings, mean of three seeds. STPT-Lite+KD+robust (proposed, selected on validation): 96.83 ± 0.69% accuracy, macro-F1 96.88. Best baseline: PointNet++ (SSG) at 97.34 ± 0.18% with 413 K parameters. The original RadHAR paper reports 90.47%. Caveat from our audit: RadHAR test recordings come from the same sessions and people as training, so this measures generalisation to new recordings, not to new people.

Possible examiner questions:

Q: *Is the difference to the best baseline significant?*

A: See the statistics slide: exact McNemar tests on the same test windows with Holm correction, per seed.

Q: *Why is RadHAR easy?*

A: Five distinct activities, same people and room in training and test.

Slide 16: Unseen people: 49.56 % on 49 actions, +17 points

Presenter: Indu Vadana M D

Primary result. Test subjects 8, 9 and 10 were never seen in training or validation. STPT-Lite+KD+robust (proposed): 49.56 ± 1.98% top-1, 78.03% top-3, macro-F1 48.80. Best baseline: PointNet 32.57 ± 4.59%. Chance is about 2%; the MiliPoint authors report 34.45% for PointNet++ under their chronological split. DGCNN and Point Transformer were compared on RadHAR only, because of their CPU cost on the much larger MiliPoint training set. Distillation alone gives 48.55 ± 4.35%; the noise-robust variant was selected on validation macro-F1.

Possible examiner questions:

Q: *Why is MiliPoint so much harder?*

A: 49 fine-grained actions, about 6 points per frame, xyz only, and new people at test time.

Q: *Why not compare to 98%?*

A: That number uses a random split; our protocol slide shows what random splitting does to our own models.

Slide 17: Gains on unseen people are significant in every seed

Presenter: Indu Vadana M D

Left: per-class confusion matrices of the proposed model (seed 0). Right: paired exact McNemar tests on the identical MiliPoint test windows (unseen people), Holm-corrected within each seed; the last column counts the seeds in which the difference is significant. Against every re-implemented baseline the difference is significant in all three seeds (largest Holm-corrected p about 1e-46). The other STPT variants (teacher, STPT-Lite with and without distillation) are close: their differences are significant only in some seeds. On RadHAR most differences are not significant, as expected for a saturated benchmark. 95% bootstrap confidence intervals (2,000 resamples) are in the paper. We do not report only the best seed.

Possible examiner questions:

Q: *Why McNemar and not a t-test?*

> **A:** Both models predict the same test windows; McNemar uses exactly the disagreements, which is the appropriate paired test for classifiers.

Slide 18: Robust training helps; false detections remain a weakness

Presenter: Murugan S

Accuracy on the full test set under each perturbation and severity; identical perturbed samples for every model. Findings. MiliPoint (right): the proposed model is the most accurate under every condition except extreme sparsity (12.5% of points kept, about one point per frame), where it falls to PointNet level; noise-robust training improves it under every perturbation by about 1 to 5 points, including jitter, which was not used in training. RadHAR (left): dropout, density and jitter up to 5 cm cost at most 2.5 points; losing half of the segments costs our model only 1.2 points. The weakness: at 20% injected false detections our model keeps 55% (39% without robust training) while PointNet with the same robust training keeps 96% and the voxel model 94%. Likely cause: outliers shift the centroid and Doppler statistics of the motion descriptor. We report this openly; a median-based descriptor is future work. Solid red is the proposed model (distillation + noise-robust training); dashed orange is the same model without noise-robust training; blue is PointNet, dashed blue PointNet with the same noise-robust training. Read the outlier and sparsity panels first: these are the conditions that hurt most. The prediction-stability metric (fraction of predictions unchanged) and per-class recall are in the result files and paper.

Possible examiner questions:

Q: *Did noise-robust training help?*

> **A:** Compare the solid red and dashed orange lines; the paper reports the exact differences per condition, including where it did not help (extreme sparsity, outliers on RadHAR).

Q: *Why not test on real noisy recordings?*

> **A:** No labelled noisy recordings exist for these datasets; controlled perturbations are the measurable alternative and are labelled as such.

Slide 19: 2.58 ms per window, 0.14 MB with INT8 weights

Presenter: Murugan S

Accuracy versus parameters and CPU latency. Proposed model: 117.5 K parameters, 36.60 MFLOPs, 2.58 ms per window on one CPU thread, 0.49 MB in FP32 and 0.14 MB with INT8 weights (INT8 accuracy 96.82%). Latency was measured on a desktop CPU thread; we did not deploy on a micro-controller.

Possible examiner questions:

Q: *Why weight-only INT8?*

A: PyTorch dynamic INT8 lost 7 points and was slower for these tiny layers; weight-only INT8 keeps accuracy and cuts storage.

Slide 20: Temporal Transformer removed: −11.9 points on unseen people

Presenter: Murugan S

Each bar removes one component of STPT-Lite (no distillation) and retrains; the bar is the change in test accuracy relative to the full model trained with the same seeds (MiliPoint: 2 seed(s), RadHAR: 2) (RadHAR full: 96.22 ± 1.43%, MiliPoint full: 45.79 ± 6.30%). RadHAR is saturated: every change there lies within the seed spread of the full model (±1.43), so RadHAR cannot rank components. MiliPoint, with unseen people and 49 actions, is the informative benchmark for the architectural components; Doppler, gating and augmentation are RadHAR-specific inputs and are tested there only. The full table with standard deviations is in the paper.

Possible examiner questions:

Q: *Which component matters most?*

A: On MiliPoint, temporal Transformer: removing it changes accuracy by -11.85 points.

Q: *Does the motion descriptor help?*

A: Not measurably: on MiliPoint removing it changes accuracy by +0.5 points on average (+2.8 and -1.8 in the two seeds). We keep it because it costs only 2.5 K parameters, but we do not claim it as an accuracy contribution.

Q: *Some RadHAR variants are better than the full model. Why keep the component?*

A: Those differences are inside the seed spread of a saturated benchmark (all models above 95%); components are judged on MiliPoint, where removing attention or the temporal Transformer costs 6.5 and 11.9 points in both seeds.

Slide 21: How the split changes the answer (MiliPoint)

Presenter: Indu Vadana M D

Same models, same code, three splits. Subject-disjoint (ours): new people at test time. Chronological (MiliPoint code): last 10% of frames, which shares subjects with training. Random windows: overlapping windows shuffled, so most test frames were seen in training. The jump from the first to the last column is leakage, not learning.

Possible examiner questions:

Q: *So which number should be reported?*

A: The subject-disjoint one, if the claim is that the system works for new people.

Slide 22: MiliPoint pre-training: +18 points with 10 % of RadHAR data

Presenter: Indu Vadana M D

We pre-train STPT-Lite on MiliPoint (xyz, harmonised axes) and fine-tune on RadHAR with 10%, 25% and 100% of the training recordings, comparing with training from scratch on the same subsets (xyz input for both, two seeds). Result: with 10% of the RadHAR recordings, pre-training raises accuracy from 61.22% to 78.98%; at 25% the gain is about one point and inside the seed spread; with all data both are equivalent. Different radar, frame rate, people, room and labels, so the tokenizer appears to learn largely sensor-independent motion features, which matters when a new home has few labelled recordings.

Possible examiner questions:

Q: *Why only xyz?*

A: MiliPoint has no Doppler or intensity; using xyz on both sides keeps the comparison clean.

Slide 23: The model integrates evidence over the whole window

Presenter: Murugan S

Left: per-class temporal attention of the [CLS] token over the 12 segments and the occlusion importance (drop in true-class probability when a segment is removed). Right: gradient saliency per point for one window per class, and a t-SNE of the learned embeddings. Attention is not proof of causation; occlusion and saliency are complementary checks.

Possible examiner questions:

Q: *Does attention explain the decision?*

A: Not on its own (Jain & Wallace 2019); we compare it with occlusion, which directly measures the effect of removing a segment.

Slide 24: Live demo runs the saved model end to end

Presenter: Murugan S

Live demo: streamlit run demo/app.py. It loads the exported checkpoint (the seed chosen by validation), parses a raw RadHAR recording or a MiliPoint sample, shows the raw and pre-processed point clouds, predicts per 2-s window with class probabilities and latency, and lets you inject each noise type and compare predictions. Bundled samples come only from the held-out test split. Invalid files are rejected with a message. Everything is computed by the real model; nothing is hard-coded.

Possible examiner questions:

Q: *Is the demo using the real model?*

A: Yes; tab 4 shows the checkpoint path and its test metrics, and any upload is processed by the same code as evaluation.

Q: *What if I upload a wrong file?*

A: The app validates format, shape, NaN and length and shows an error instead of crashing.

Slide 25: Limitations and future work

Presenter: Murugan S

Be explicit with examiners. RadHAR cannot test new people or rooms. MiliPoint is xyz-only and has three test subjects. Robustness uses synthetic perturbations, not a radar simulator. Latency is measured on a desktop CPU thread. No fall labels, single person only. Future work follows directly from these limitations.

Possible examiner questions:

Q: *What would you do with more time?*

> **A:** Leave-one-subject-out and cross-room evaluation, on-device deployment, and self-supervised pre-training on the many unlabelled frames.

Slide 26: Conclusion

Presenter: Murugan S

Summarise in three sentences: what we built (a factorised point transformer for sparse radar), what we showed (competitive accuracy at very low cost, honest numbers on unseen people, robustness profile), and what we learned (evaluation protocol matters as much as the model). Thank the audience and invite questions.

Possible examiner questions:

Q: *What is the single most important finding?*

> **A:** Under leakage-free evaluation a 118 K-parameter model is competitive, and random splits inflate MiliPoint accuracy dramatically.

Slide 27: References (selected)

Presenter: Abarajitha S K

Selected references; the paper lists all of them in IEEE format.

Possible examiner questions:

Q: *Are all references real?*

> **A:** Yes, each was verified (authors, venue, year) before citing.